HOMEWORK 2

1. Let d be an integer such that $d \neq 1$ and d is not divisible by the square of a prime integer. Let $R = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ if $d \equiv 1$ modulo 4 and $R = \mathbb{Z}[\sqrt{d}]$ otherwise. Prove that R is a Dedekind ring. (Hint: In order to prove that R is normal, let $x + y\sqrt{d} \in \mathbb{Q}(\sqrt{d})$ with $x, y \in \mathbb{Q}$ be integral over $\mathbb{Z}$. Show that $2x$ and $x^2 - dy^2$ are in $\mathbb{Z}$ and deduce that $x + y\sqrt{d} \in R$.)

2. Let p be prime integer. Prove that if $p \equiv 3$ modulo 4, then pR is a prime ideal in the ring of Gauss integers $R = \mathbb{Z}[i]$. Prove that if $p \equiv 1$ modulo 4, then pR is a product of two distinct prime ideals of R. Find a prime ideal P in R such that $2R = P^2$.

3. Let $R = \mathbb{Z}[\sqrt{-5}]$. Factor $14R$ into the product of prime ideals.

4. Is $\mathbb{Z}[\sqrt{5}]$ a Dedekind ring?

5. Let $R := \mathbb{Q}[x, y]/(x^2 + y^2 - 1)\mathbb{Q}[x, y]$, i.e., R is the ring of polynomial functions on the unit circle over $\mathbb{Q}$.
(i) Prove that R is a domain with quotient field $K = \mathbb{Q}(x)(\sqrt{1 - x^2})$ that is a quadratic extension of $\mathbb{Q}(x)$.
(ii) Show that every element of K can be uniquely written in the form $f + g \cdot y$, where $f, g \in \mathbb{Q}(x)$.
(iii) Prove that R is a Dedekind ring. (Hint: In order to prove that R is normal, show that $f + g \cdot y$ in K with $f, g \in \mathbb{Q}(x)$ is integral over $\mathbb{Q}[x]$ if and only if f and g are in $\mathbb{Q}[x]$.)

6. Factor the ideal yR in the ring $R := \mathbb{Q}[x, y]/(x^2 + y^2 - 1)\mathbb{Q}[x, y]$ into the product of prime ideals. (Hint: Divide yR by $(1 - x)R + yR$.)

7. Let R be a Dedekind ring and $S \subset R$ a multiplicative subset. Prove that the localization $S^{-1}R$ is also a Dedekind ring.

~~8.~~ Prove that for every two ideals A and B of a Dedekind ring one has $AB = (A + B)(A \cap B)$.

9. Prove that the ring of rational functions $f/g \in F(x)$ over a field F with $\deg(f) \leq \deg(g)$ is a DVR.

~~10.~~ Prove that a Noetherian integral domain R is a Dedekind ring if and only if the localization $R_P := S^{-1}R$, where $S = R - P$, at every prime ideal P is a DVR.